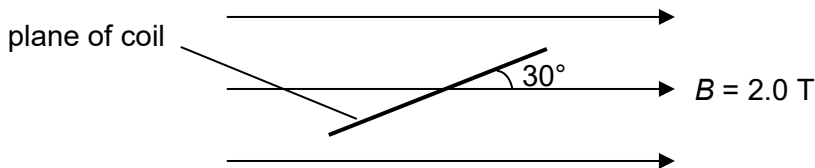


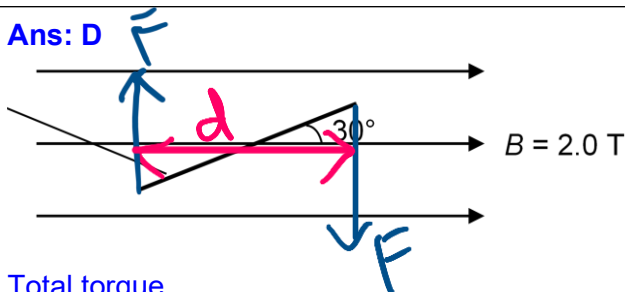
- 24 A square coil of 10 turns with sides of 5.0 cm is placed so that the plane of the coil makes an angle of 30° with the direction of a uniform magnetic field B of flux density 2.0 T. A current of 15 A is passed through the coil.



What is the magnitude of the torque acting on the square coil?

- A 0.19 N m B 0.33 N m C 0.38 N m D 0.65 N m

Ans: D



Total torque,

$$\begin{aligned}
 \tau &= F \times d \\
 &= NBILd \\
 &= 10 \times 2.0 \times 15 \times \frac{5.0}{100} \times \frac{5.0}{100} \cos 30^\circ \\
 &\approx 0.65 \text{ Nm}
 \end{aligned}$$

- 25 A flat circular coil of 120 turns, each of area 0.070 m^2 is placed with its axis parallel to a uniform